

MEALVERSE

DISCOVER, MASTER, AND SHARE CULINARY ARTS

ICT 1209 – Web Technologies

Mini Project:

Interactive Web Application Development

Group Members

H.M.T.L. Bandara

Index No. : 2703

H.P.P.B.K. Dharmarathna

Index No. : 2715

Contents

1. Project Overview & Objective

- 1.1 Introduction
- 1.2 Purpose
- 1.3 Target Audience
- 1.4 Key Features

2. System Design

- 2.1 Database Schema (Relational Architecture)
- 2.2 Directory & File Structure

3. Implementation

- 3.1 Dynamic Landing Hub & Live Platform Analytics
- 3.2 Recipe Creation & In-Depth Culinary View
- 3.3 Culinary Marketplace & Shopping Cart Workflow
- 3.4 Master Chef Directory & Engagement
- 3.5 Contact & Technical Support System

4. Challenges & Solutions

- 4.1 Reusable DOM Components for Layout Consistency
- 4.2 Client-Side Multi-Category & Real-Time Filtering
- 4.3 Dynamic Form Row Generation in Recipe Submission
- 4.4 Multi-Device Grid Responsiveness & Media Optimization

5. Individual Contribution

6. Setup & Execution Instructions

- 6.1 Server Environment Setup
- 6.2 Application Directory Placement
- 6.3 Database Import
- 6.4 Running the Web Application

7. References

1. Project Overview & Objective

1.1 Introduction

MealVerse is a modern, responsive web application designed for discovering, sharing, and mastering culinary arts. The platform connects everyday cooking enthusiasts, aspiring food creators, and professional culinary masters into a single interactive ecosystem.

1.2 Purpose

The core objective of MealVerse is to modernize recipe management and food exploration by replacing static cookbooks with an interactive digital workspace. The platform enables users to:

- Explore international recipes categorized by dietary profile and cuisine type.
- Publish personalized step-by-step recipes with modular ingredient lists.
- Discover curated kitchen tools, cookbooks, and gourmet ingredients via an integrated store.
- Connect with world-class chefs through ratings, verified stats, and direct social interaction.

1.3 Target Audience

- **Home Cooks & Beginners:** Individuals seeking clear preparation times, calorie counts, and step-by-step instructions.
- **Professional Chefs & Content Creators:** Culinary experts who wish to publish signature recipes and cultivate a follower base.
- **Culinary Shoppers:** Users searching for specialized kitchenware, rare ingredients, and curated cookbooks.

1.4 Key Features

- **Multi-Attribute Search & Filter:** Instant filtering of recipes across multiple categories like Main Course, Appetizers, Desserts, Soups, Salads, and Beverages.
- **Modular Recipe Publishing Portal:** A structured form with designated panels for basic details, photo uploads, dynamic ingredient additions, and numbered instructions.
- **Culinary Marketplace & Cart:** A product catalog with client-side category switching, quantity toggles, discount summaries, and checkout pricing.
- **Chef Directory:** Profile cards highlighting master chefs with verified badges, follower counts, and overall user ratings.
- **Detailed Recipe Viewer:** Comprehensive presentation displaying preparation times, cooking times, calorie counts, ingredient portions, and step-by-step guidance.
- **Direct Feedback Gateway:** A responsive contact interface for inquiries, complete with location information and form validation.

2. System Design

2.1 Database Schema (Relational Architecture)

The backend database relies on a normalized structure designed to ensure transactional consistency across user activities, catalog records, and shopping orders.

mealverse_db users id : int(11) name : varchar(100) email : varchar(150) password : varchar(255) created_at : timestamp	mealverse_db recipes id : int(11) title : varchar(255) chef : varchar(100) image : varchar(255) cuisine : varchar(100) category : varchar(100) instructions : text created_at : timestamp	mealverse_db marketplace_items id : int(11) name : varchar(255) category : varchar(100) description : text price : decimal(10,2) image : varchar(255) stock : int(11) created_at : timestamp
---	--	---

Figure 1 Some of Database Schemas

- **users:** Stores authenticated account details, email addresses, and roles.
- **recipes:** Stores the primary metadata for submitted recipes, including cuisine tags, servicing size, and total calories.
- **products / marketplace:** Contains merchandise listings, pricing, review counts, discount tags, and product categories.
- **cart items:** Tracks active cart selections, item quantities, and per-unit prices prior to checkout.
- **messages:** Stores user feedback, questions, and communication sent through the contact gateway.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	id	int(11)			No	None		AUTO_INCREMENT	Change Drop More
2	name	varchar(100)	utf8mb4_general_ci		No	None			Change Drop More
3	email	varchar(150)	utf8mb4_general_ci		No	None			Change Drop More
4	password	varchar(255)	utf8mb4_general_ci		No	None			Change Drop More
5	created_at	timestamp			No	current_timestamp()			Change Drop More

Figure 2 User Database

2.2 Directory & File Structure

The project files are arranged in a modular layout separating core HTML templates, presentation styles, script modules, and static assets:

```
mealverse/
|
├── css/
|   ├── style.css      # Typography, layout framing, color palette, custom cards
|   ├── responsive.css # Breakpoint definitions and media query adjustments
|   └── images/        # Badges, culinary thumbnails, and chef avatars
|
├── js/
|   ├── nav.js         # Dynamic DOM injector for site-wide navigation
|   └── footer.js      # Site-wide footer injector script
|
├── addrecipe.html     # Multi-stage recipe submission form
├── cart.html          # Shopping cart overview and pricing summary
├── chefs.html         # Culinary artist directory and chef stats
├── contact.html       # Inquiry form and support contact details
├── index.html         # Landing hub with featured items and platform metrics
├── marketplace.html   # Kitchenware and ingredient storefront
├── recipes.html       # Recipe directory with category filter tabs
├── viewrecipe.html    # Individual recipe view with full cooking breakdown
|
├── database.sql       # Relational SQL export for table setup
└── README.md         # Installation and operational documentation
```

3. Implementation

3.1 Dynamic Landing Hub & Live Platform Analytics

The main landing page introduces the web platform with a search bar and platform metrics:

- **Hero Search Integration:** Allows visitors to filter through recipes, chefs, or ingredients from a single input.
- **Platform Metrics:** Highlights engagement through metric counters
- **Featured & Trending Grids:** Displays recipe cards marked with specialized tags such as "FREE", "PREMIUM", or region indicators like "Sri Lankan" and "French".

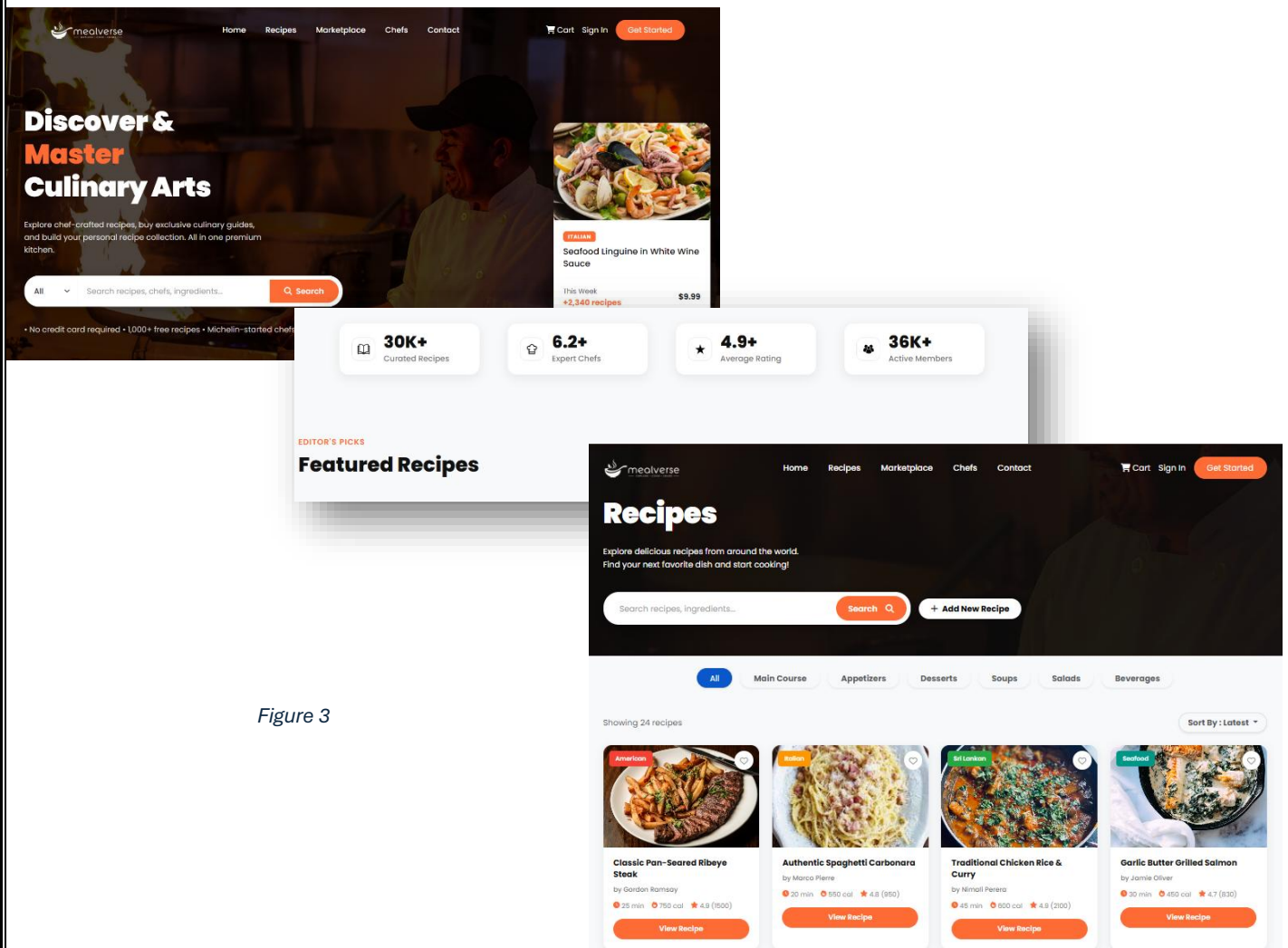


Figure 3

3.2 Recipe Creation & In-Depth Culinary View

- **Creation Portal:** Organizes user input into four distinct sections: Basic Information, Recipe Image Upload, Dynamic Ingredients Input, and Step-by-Step Instructions.
- **Detail Display:** Features an overlay hero image with author credentials, preparation stats an ingredients list, and numbered execution step

SHARE YOUR CREATIONS
Add New Recipe
Turn your favorite dish into inspiration for a global food community.

01 **Basic Information** Tell us about your recipe

Recipe Title

Short Description

CUISINE CATEGORY SERVINGS

Select cuisine Select category Servings

02 **Recipe Image** Add a mouthwatering photo

Upload Recipe Image
Drag and drop an image here, or click to browse
JPG, PNG up to 5MB

03 **Ingredients** List what's needed

Ingredient name Quantity

+ Add Ingredient

04 **Instructions** Share the cooking steps

Figure 4 Add Recipe

3.3 Culinary Marketplace & Shopping Cart Workflow

- **Product Catalog:** Categorizes items into Kitchen Tools, Ingredients, and Cookbooks, allowing instant category filtering via button triggers.
- **Order Processing UI:** Summarizes selected products, quantity adjustments, item removals, discount vouchers, taxes, and final order calculations.

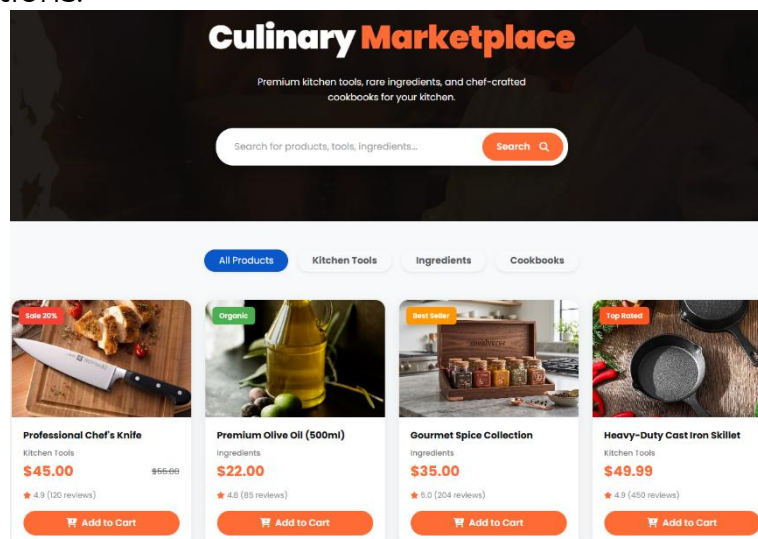


Figure 5 Marketplace

3.4 Master Chef Directory & Engagement

- **Chef Index:** Highlights culinary experts using profile cards that show specialties, verified tags, follower numbers, total recipes, and ratings.
- **Community Invitation:** Includes an interactive call-to-action banner inviting passionate home cooks and professionals to apply as master chefs.

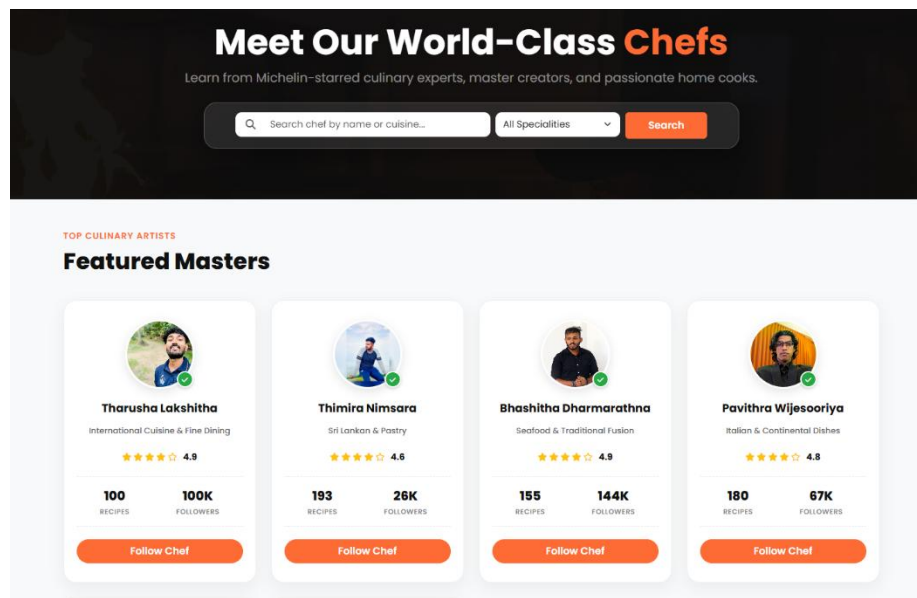


Figure 6 Chefs

3.5 Contact & Technical Support System

- **Communication Gateway:** Displays university contact credentials alongside a validated form for direct message transmission.

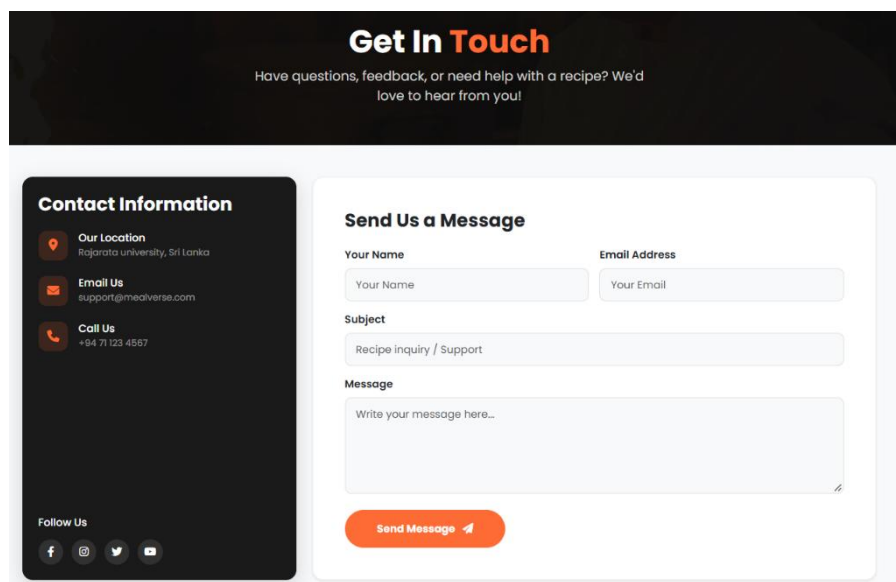


Figure 7 Contact

4. Challenges & Solutions

4.1 Reusable DOM Components for Layout Consistency

- **Problem:** Manually duplicating navigation bars and footer markup across multiple pages caused maintenance overhead and inconsistent links.
- **Solution:** Modular JavaScript injection scripts (nav.js and footer.js) were implemented to dynamically render uniform headers and footers into #navbar-placeholder and #footer-placeholder elements.

4.2 Client-Side Multi-Category & Real-Time Filtering

- **Problem:** Switching categories on the recipe and marketplace pages caused noticeable latency when handled with full page reloads.
- **Solution:** Vanilla JavaScript DOM event listeners were paired with data-category and data-filter HTML5 attributes, enabling instant visibility toggling (display: block / none) directly in the browser.

4.3 Dynamic Form Row Generation in Recipe Submission

- **Problem:** Supporting variable recipe lengths required adding and deleting ingredient fields and instructions dynamically without breaking the form layout.
- **Solution:** JavaScript functions were added to listen for clicks on + Add Ingredient and + Add Another Step, programmatically creating and appending input elements to the parent form container.

4.4 Multi-Device Grid Responsiveness & Media Optimization

- **Problem:** Complex 4-column card grids on desktop screens looked crowded and misaligned on smaller mobile viewports.

- **Solution:** Standardized on Bootstrap 5.3's responsive grid system (combining col-12, col-md-6, and col-lg-3 breakpoints) and added custom media queries in responsive.css to handle consistent layout reflows across devices.

5. Individual Contribution

The development of the MealVerse platform was executed through an equal and collaborative team distribution (50% / 50% split). Both members engaged in full-stack implementation, sharing responsibilities across user interface design, client-side functionality, server endpoints, and data storage:

Member 1: Full-Stack Web Developer & Logic Engineer

Name: H.M.T.L.Bandara

Index Number: 2703

- Marketplace & E-Commerce Module (marketplace.html, get_marketplace.php): Implemented the culinary storefront with instant category tab switching (Tools, Ingredients, Cookbooks) and developed the backend endpoint to fetch catalog inventory.
- Shopping Cart & Checkout Flow (cart.html, cart.php, cart_api.php, checkout.php): Developed the cart management module, price calculations, session-based state management, and transactional order submissions with stock updates.
- Master Chef Directory (chefs.html): Designed the culinary artists directory, presenting verified chef profile badges, follower metrics, review ratings, and the "Become a Chef" call-to-action banner.
- Global Navigation Component (nav.js): Engineered the reusable client-side DOM injector script to render uniform navigation bars across every webpage without duplicate template code.
- Inquiry & Feedback Handling (contact.html, process_contact.php): Constructed the contact interface, input constraint validations, and backend message storage workflow.

Member 2: Full-Stack Web Developer & UI Designer

Name: H.P.P.B.K.Dharmarathna

Index Number: 2715

- Landing Hub & Platform Analytics (index.html): Developed the main landing page, integrating the top hero showcase, multi-attribute search dropdown, and dynamic platform metric counters (Curated Recipes, Master Chefs, and Active Members).
- Recipe Explorer & Filtering Engine (recipes.html, get_recipes.php): Built the multi-cuisine recipe directory, integrating real-time category filtering for dishes (Main Course, Desserts, Appetizers) and backend retrieval queries.
- Recipe Creation Portal (addrecipe.html, addrecipe.php, add_recipe.php): Implemented the recipe authoring interface with dynamic row insertion/removal for ingredients and cooking steps, as well as file upload handling for recipe imagery.
- Single Recipe Detail Template (viewrecipe.html): Engineered the detailed recipe overview displaying preparation timing, calorie counts, bulleted ingredient portions, and numbered step-by-step instructions.
- Global Footer & Responsive Optimization (footer.js, responsive.css): Built the dynamic footer injector and refined custom media query breakpoints to guarantee mobile and tablet responsiveness.

6. Setup & Execution Instructions

6.1 Server Environment Setup

1. Download and install a local web stack, such as **XAMPP** or **WampServer**.
2. Open the control panel and start both the **Apache** and **MySQL** services.

6.2 Application Directory Placement

1. Extract the project source archive.
2. Copy the root mealverse folder into the web server document root directory (e.g., C:\xampp\htdocs\mealverse\).

6.3 Database Import

1. Open a web browser and navigate to <http://localhost/phpmyadmin>.
2. Create a new database named `mealverse_db` using the `utf8mb4_general_ci` collation.
3. Click the **Import** tab, select the provided `database.sql` file from the project directory, and click **Import** to generate the schema.

6.4 Running the Web Application

1. In the browser, navigate to:

<http://localhost/mealverse/index.html>

2. Explore the live features, including the recipe catalog, marketplace filter tabs, cart summary, and chef showcase.

7. References

✓ **Bootstrap 5 Framework:**

Bootstrap Core Team. (2024). *Bootstrap 5.3 Official Documentation – Responsive Grid, Components, and Utilities*. Available at: <https://getbootstrap.com/docs/5.3/getting-started/introduction/>

✓ **MDN Web Docs (JavaScript, DOM & HTML5 APIs):**

Mozilla Developer Network. (2024). *Web Technology for Developers: Document Object Model (DOM), Event Listeners, and HTML5 Semantic Elements*. Available at: <https://developer.mozilla.org/en-US/>

✓ **PHP Official Documentation:**

The PHP Group. (2024). *PHP Manual: MySQLi Extension, Prepared Statements, File Uploads (\$_FILES), and Session Handling (\$_SESSION)*. Available at: <https://www.php.net/manual/en/>

✓ **MySQL 8.0 Reference Manual:**

Oracle Corporation. (2024). *MySQL 8.0 Reference Manual – Relational Database Design, Foreign Keys, InnoDB Storage Engine, and ACID Transactions*. Available at: <https://dev.mysql.com/doc/refman/8.0/en/>

✓ **Apache Friends XAMPP Server Environment:**

Apache Friends. (2024). *XAMPP Apache + MariaDB + PHP + Perl Development Environment Documentation*. Available at: <https://www.apachefriends.org/>

✓ **Google Fonts & Typography (Poppins):**

Google Fonts. (2024). *Poppins Typeface Family designed by Indian Type Foundry*. Available at: <https://fonts.google.com/specimen/Poppins>

✓ **Google Material Symbols & Icons:**

Google Fonts. (2024). *Material Symbols and Icons Guide*. Available at: <https://fonts.google.com/icons>

✓ **Font Awesome Icon Library:**

Fonticons, Inc. (2024). *Font Awesome 6 Free SVG & Web Icons*. Available at: <https://fontawesome.com/>

✓ **Unsplash Visual Media & Photography Assets:**

Unsplash Inc. (2024). *High-Resolution Royalty-Free Food and Culinary Photography Archive*. Available at: <https://unsplash.com/>